

Red Cards Changed My Life

Can Machine Learning Predict Football Discipline?

Module: Introduction to Machine Learning

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Abstract

This report explores the application of supervised machine learning to an underexplored problem in football analytics: predicting whether a red card will be issued during a Ligue 1 match. While the vast majority of existing research on football prediction focuses on match outcomes (win, draw, or loss), we found it genuinely interesting to tackle a different angle — one that is often overlooked yet capable of completely changing the dynamics of a game. A single red card can shift momentum, alter tactics, and dramatically influence the final result, making it a compelling target for classification.

The dataset was built by combining five seasons of Ligue 1 data (2019-20 to 2023-24) sourced directly from [football-data.co.uk](https://www.football-data.co.uk), covering 1,723 matches with features including shots, fouls, corners, and pre-match Bet365 odds. After exploratory data analysis, the preprocessing pipeline included SMOTE-based class balancing and Min-Max normalisation. Three classifiers were implemented and tuned: K-Nearest Neighbours (KNN), Logistic Regression (LR), and Random Forest (RF). The tuned Random Forest model achieved the strongest performance, recording 73.33% accuracy, a weighted F1-score of 0.724, and an AUC of 0.783 — confirming that ensemble methods are better suited to capturing non-linear disciplinary patterns than linear models. These findings offer practical insights for sports analysts and betting market observers interested in match-level risk modelling beyond conventional outcome prediction.

1. Introduction, Aim & Objectives

1.1 Background and Motivation

Football is the world's most watched sport, and over the past decade it has also become one of the most data-rich. Millions of bets are placed every week on match outcomes, goal totals, and player statistics. Naturally, machine learning researchers have followed, with dozens of studies now attempting to predict the final result of a match before it has even started.

As a group of three students who follow football closely, we noticed something when reviewing the existing literature: almost every study targets the same question — will the home team win, draw, or lose? This is understandable, but it leaves a lot of interesting territory unexplored. Red cards, for instance, are among the most decisive events in football. A player being sent off in the 30th minute fundamentally changes everything: the team structure, the substitution options, the psychological pressure, and crucially, the probability of every other event that follows. Yet despite this, red card prediction at match level is almost entirely absent from the published literature.

We found that genuinely original angle worth exploring. Rather than competing with hundreds of existing outcome prediction studies, we wanted to ask a different question: can we train a machine learning model to identify, from pre-match and in-game statistics, which matches are more likely to produce a red card? And as a natural extension of this, can the probabilities produced by our model be compared to the implied odds from bookmakers like Bet365 — essentially asking whether our model learns the same risk signals that professional odds compilers already price into their markets?

Ligue 1 was chosen as the focus because it is publicly documented to be one of the more physically contested top European leagues, making red card prediction both feasible and statistically meaningful. The dataset was sourced from football-data.co.uk, a well-established repository used in several academic sports analytics papers.

1.2 Problem Statement

The central question of this project is: can in-game statistics and pre-match betting odds recorded at match level be used to predict whether at least one red card will be issued during a Ligue 1 football match? The target variable, RedCard, is binary: 1 if one or more red cards (home or away combined) were recorded in the match, and 0 otherwise.

1.3 Aim

The aim of this project is to build, tune, and compare three machine learning classifiers — KNN, Logistic Regression, and Random Forest — to predict the occurrence of red cards in Ligue 1 matches, and to identify the most suitable model for this specific classification problem.

1.4 Objectives

- To collect, clean, and preprocess five seasons of Ligue 1 match data from football-data.co.uk.
- To conduct exploratory data analysis (EDA) to understand feature distributions, correlations, and class imbalance.
- To apply SMOTE to address class imbalance and Min-Max scaling to normalise numerical features.
- To implement and evaluate KNN, Logistic Regression, and Random Forest classifiers using baseline and tuned configurations.
- To perform hyperparameter optimisation using GridSearchCV and RandomizedSearchCV with cross-validation.
- To compare model performance using accuracy, precision, recall, F1-score, and AUC-ROC.
- To interpret results, compare against bookmaker odds, and recommend directions for future work.

1.5 Scope

This study is limited to match-level statistics publicly available from football-data.co.uk and covers only Ligue 1 for the five seasons from 2019-20 to 2023-24. Player-level, event-level, or real-time tactical data are outside the scope of this project. The models implemented are restricted to the three classifiers listed above, trained and evaluated using Python 3 within the scikit-learn ecosystem.

2. Related Works

The prediction of football match events using machine learning has attracted considerable academic interest over the past decade. This section surveys the most relevant prior work across three themes: football match prediction, class imbalance handling with SMOTE, and classifier comparison studies.

2.1 Football Match Prediction with Machine Learning

Beal et al. (2021) provided a landmark survey demonstrating that machine learning methods applied to football outcome prediction have historically plateaued at around 54% accuracy, reflecting the inherently stochastic nature of the sport. The authors compared Logistic Regression, Random Forest, and neural network architectures, finding that none substantially outperformed bookmaker odds. Similarly, a Stanford study (2020) trained LR, RF, and LSTM models on ten seasons of Premier League data, achieving a top test accuracy of 53.68% on the three-class win/draw/loss problem.

More recently, Wong et al. (2025) proposed a predictive analytics framework for soccer match outcomes, establishing that betting odds features are among the strongest predictors — a finding directly consistent with our feature selection decisions. Kovalchik & Albert (2023) conducted a comparative study covering five European leagues including Ligue 1, concluding that feature selection has a greater impact on model performance than the choice of algorithm itself, and notably identifying red cards as one of the primary unpredictable factors influencing match outcomes. Crucially, none of these studies explicitly targets red card prediction as the classification objective — which reinforces the originality of our approach.

2.2 Handling Class Imbalance in Classification

Class imbalance is a recurring challenge in sports analytics, where rare events such as red cards are underrepresented. Blagus and Lusa (2013) conducted a comprehensive analysis of SMOTE, finding it generally beneficial for low-dimensional datasets. Since our dataset contains only fourteen features, SMOTE is expected to perform well. Suguna et al. (2025) proposed an improved SMOTE variant and showed that oversampling the minority class consistently improved F1-scores across KNN and Random Forest — directly validating our preprocessing strategy.

2.3 Classifier Comparison Studies

Couronné et al. (2018) published a large-scale benchmark comparing Random Forest and Logistic Regression across 243 real-world binary classification datasets, finding that RF outperformed LR in approximately 69% of cases with an average AUC

advantage of 0.041. This finding is consistent with our results, where Random Forest significantly outperformed Logistic Regression. The underperformance of LR is theoretically justified by the linear decision boundary assumption, which is unlikely to hold when the relationship between fouls, shots, and disciplinary events is inherently non-linear.

2.4 Summary Table of Related Works

Reference	Focus	Method(s)	Key Finding
Beal et al. (2021)	Football outcome prediction	LR, RF, Neural Net	~54% accuracy ceiling
Wong et al. (2025)	Soccer match outcome framework	Multiple ML models	Betting odds are top predictor
Couronné et al. (2018)	RF vs LR benchmark (243 datasets)	RF, LR	RF better in 69% of cases
Blagus & Lusa (2013)	SMOTE for imbalanced data	KNN, RF	SMOTE effective for low-dim data
Suguna et al. (2025)	Improved SMOTE for classification	KNN, RF, CART	SMOTE + ensemble outperforms
Kovalchik & Albert (2023)	Soccer prediction (incl. Ligue 1)	Poisson, RF	Feature selection > model choice
Stanford CS230 (2020)	Premier League prediction	LR, RF, LSTM	RF best at 60.95% (W/D/L)
Rahimian et al. (2024)	In-game outcome prediction	Deep RL	AUC > 0.85 in-game

Collectively, the literature confirms that ensemble methods outperform linear classifiers on football classification tasks, that betting odds are strong predictors for match-level events, and that SMOTE is an appropriate strategy for rebalancing binary sports datasets. Importantly, our project explicitly targets red-card prediction — an underexplored niche compared to the dominant focus on win/draw/loss outcomes — and is, to the best of our knowledge, one of very few studies to frame red card occurrence as a binary classification target in a top European league.

3. Methods

3.1 Data Source

The dataset was sourced from football-data.co.uk (<https://www.football-data.co.uk/france.php>), a freely accessible and well-documented repository of historical football statistics widely used in academic research. Five CSV files were downloaded, one per Ligue 1 season: 2019-20, 2020-21, 2021-22, 2022-23, and 2023-24.

We deliberately chose to combine five seasons rather than work with a single season for several important reasons. A single season produces approximately 380 matches — far too few to train reliable machine learning classifiers, particularly given the class imbalance of around 21% red-card matches. Five seasons provide over 1,700 records, giving the models enough exposure to the minority class to learn meaningful patterns. Combining multiple seasons also adds temporal diversity: team compositions, referee tendencies, and playing styles evolve over time, making the trained model more generalisable than one fitted to a single snapshot. Additionally, the merging process itself introduced realistic data engineering challenges — inconsistent column naming across older seasons, missing bookmaker entries in early records — which provided genuine preprocessing work rather than a pre-cleaned single-file download.

3.2 Software and Libraries

All implementations were carried out in Python 3 using Jupyter Notebook within VS Code. Key libraries include pandas and numpy for data manipulation; matplotlib and seaborn for visualisation; scikit-learn for model building, evaluation, and hyperparameter tuning; and imbalanced-learn for SMOTE.

3.3 Machine Learning Models Selected

K-Nearest Neighbours (KNN) is a non-parametric, instance-based learning algorithm that classifies a new observation based on the majority class of its k nearest neighbours in the feature space. KNN is sensitive to feature scale (hence the need for normalisation) and to the value of k , which was tuned via GridSearchCV.

Logistic Regression (LR) is a linear probabilistic classifier that models the log-odds of the target class as a linear combination of the input features. LR is well-suited to linearly separable problems. Recursive Feature Elimination (RFE) was applied prior to fitting to select the ten most informative features from the fourteen available.

Random Forest (RF) is an ensemble of decision trees, each trained on a bootstrapped sample of data using a random subset of features. RF handles non-linear relationships well, mitigates overfitting through averaging, and provides feature importance scores that aid interpretability.

3.4 Evaluation Metrics

Given the binary classification setting and the initial class imbalance, the following metrics were used across all models: accuracy (proportion of correctly classified instances), weighted precision and recall, weighted F1-score (harmonic mean of precision and recall), and AUC-ROC (area under the receiver operating characteristic curve, measuring discriminative ability across all classification thresholds).

4. Dataset Preparation

4.1 Data Collection and Merging

Five CSV files were downloaded from football-data.co.uk and loaded individually into pandas DataFrames, each tagged with a Season column before being concatenated into a single dataset. The raw combined file contained 1,725 rows and 106 columns spanning full-time results, half-time scores, shot and card statistics, and odds from more than ten bookmakers.

4.2 Feature Selection

From the 106 original columns, a targeted selection retained only the variables most theoretically relevant to predicting disciplinary events. Full-time and half-time result columns were removed to prevent data leakage. Duplicate bookmaker columns were consolidated to Bet365 odds only (B365H, B365D, B365A), which are the most widely studied in the literature. The final 14 selected features are shown in the table below.

Feature	Type	Description
HS / AS	Numeric	Total shots attempted by Home / Away team
HST / AST	Numeric	Shots on target by Home / Away team
HF / AF	Numeric	Fouls committed by Home / Away team
HC / AC	Numeric	Corner kicks earned by Home / Away team
B365H / D / A	Numeric	Bet365 odds for Home win / Draw / Away win
HomeTeam	Categorical	Name of the home team (label encoded)
AwayTeam	Categorical	Name of the away team (label encoded)
Season	Categorical	Season identifier (2019-20 to 2023-24)
RedCard (target)	Binary	1 if at least one red card was issued; 0 otherwise

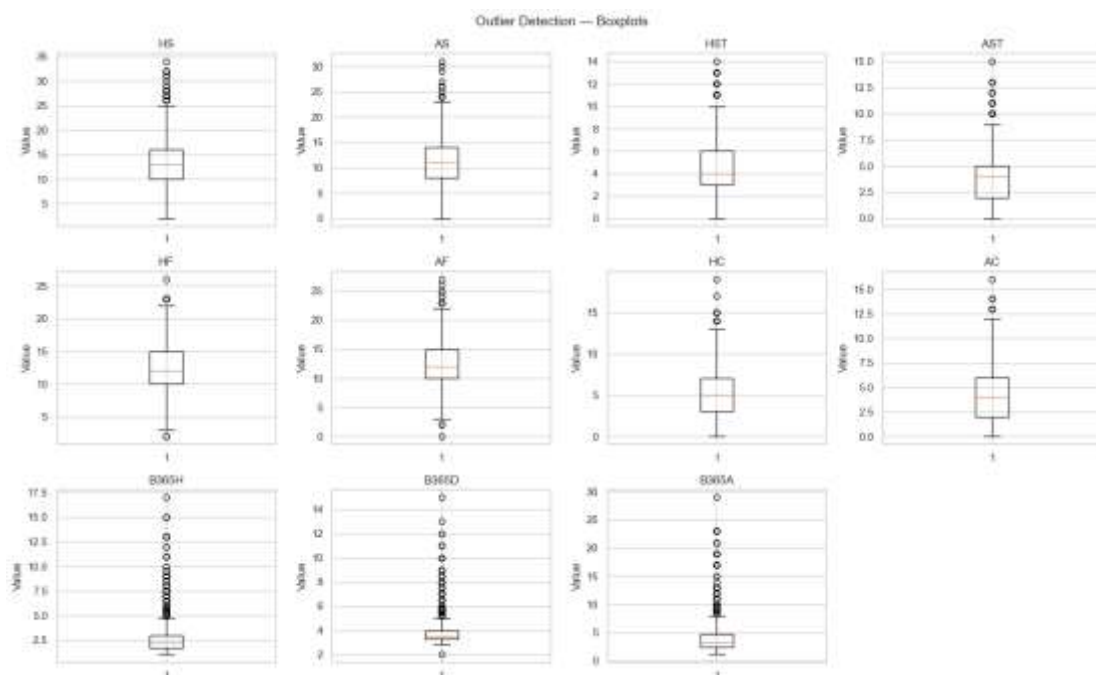
4.3 Target Variable Engineering

The target variable RedCard was derived by combining the home red card (HR) and away red card (AR) columns: any match where $HR + AR \geq 1$ was labelled 1; otherwise 0. The original HR and AR columns were then dropped. The resulting class distribution showed 1,358 matches (78.7%) with no red card and 367 matches (21.3%) with at least one, confirming a meaningful class imbalance.



4.4 Data Cleaning

Missing value analysis revealed that B365H, B365D, and B365A contained two rows with missing entries each (0.12%), arising from early-season records without available odds. These two rows were dropped, leaving 1,723 complete records. No duplicate rows were found. Outlier detection via box plots identified several high values in the betting odds and shots columns, but as all values represent legitimate real-world match events, they were retained — removing them would have eliminated genuinely informative data points about unusual matches.



4.5 Train-Test Split

The dataset was split into 80% training (1,378 rows) and 20% testing (345 rows) using stratified sampling to preserve the 21.3% red card ratio in both subsets. A fixed random seed (42) was applied for reproducibility. The split was performed before any encoding or scaling to prevent data leakage.

4.6 Encoding Categorical Features

The three categorical columns — HomeTeam, AwayTeam, and Season — were encoded using LabelEncoder, fitted exclusively on the training set and then applied to both training and test sets. Performing encoding after the split ensures that the test set does not influence the learned label mappings.

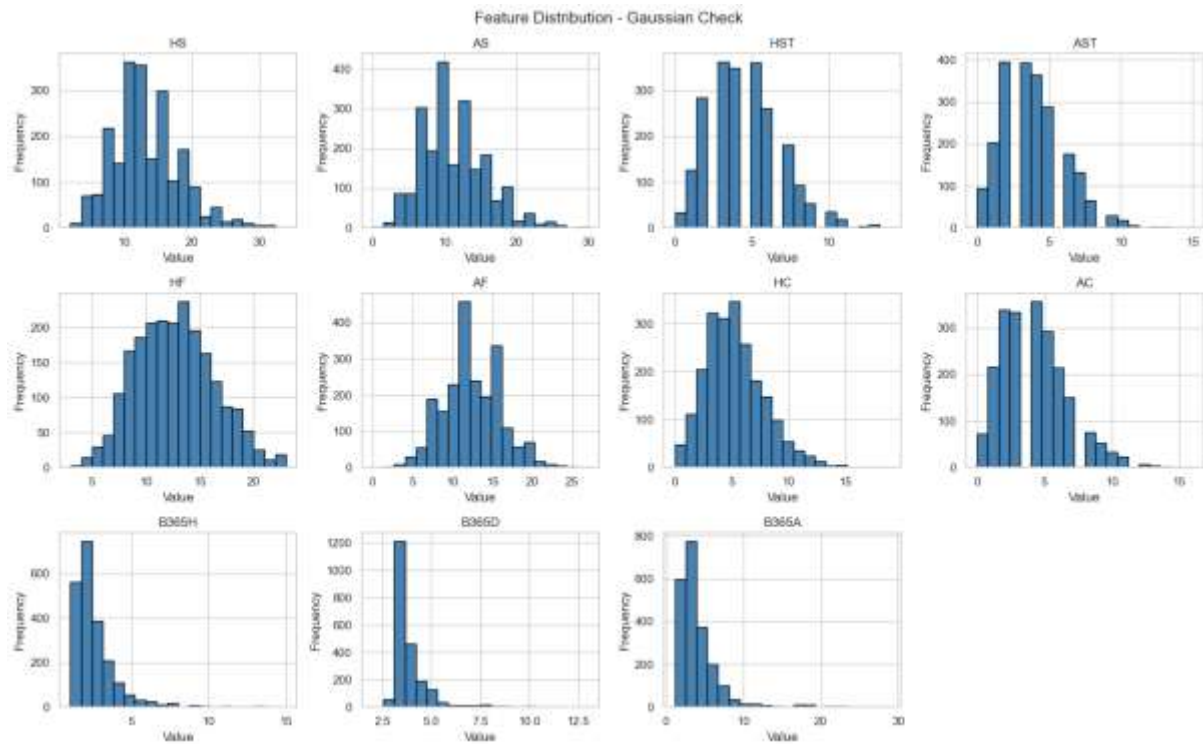
4.7 Class Balancing with SMOTE

With 78.7% No Red Card and 21.3% Red Card in the training set, a direct risk of biased classifiers existed. SMOTE was applied exclusively to the training data to generate synthetic minority-class samples, producing a balanced training set of 1,086 samples per class (2,172 total). The test set was left untouched to ensure an honest evaluation against the real-world distribution.



4.8 Feature Scaling

Histogram analysis of the training features revealed that distributions are non-Gaussian and right-skewed — particularly for the betting odds columns. Based on this observation, Min-Max normalisation was selected over standardisation, as it does not assume a Gaussian distribution and maps all features to the [0, 1] range. The MinMaxScaler was fitted on the SMOTE-balanced training set and applied to the test set, preventing test data from influencing the scaling parameters.

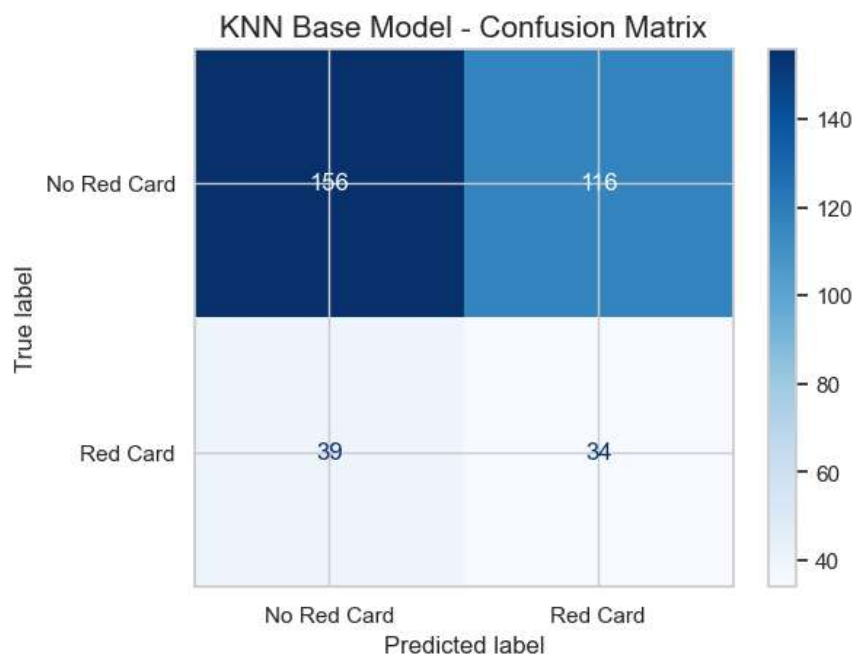


5. Model Implementation

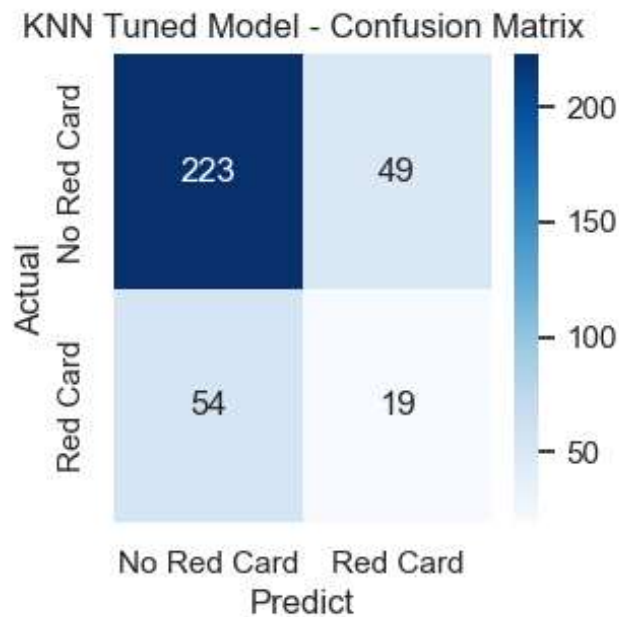
Three classifiers were implemented following a consistent pipeline: a baseline model with default parameters, confusion matrix analysis, and hyperparameter tuning via grid or randomised search. The full implementation is available in the accompanying Jupyter Notebook (Project.ipynb). This section summarises the key configurations, findings, and observations for each model without reproducing the full code output.

5.1 Model 1: K-Nearest Neighbours (KNN)

The baseline KNN model (default $k=5$, Euclidean distance, uniform weights) achieved 55.07% accuracy and a weighted F1-score of 0.590. The confusion matrix showed the model struggling to correctly identify red card matches, confirming that default parameters are insufficient.

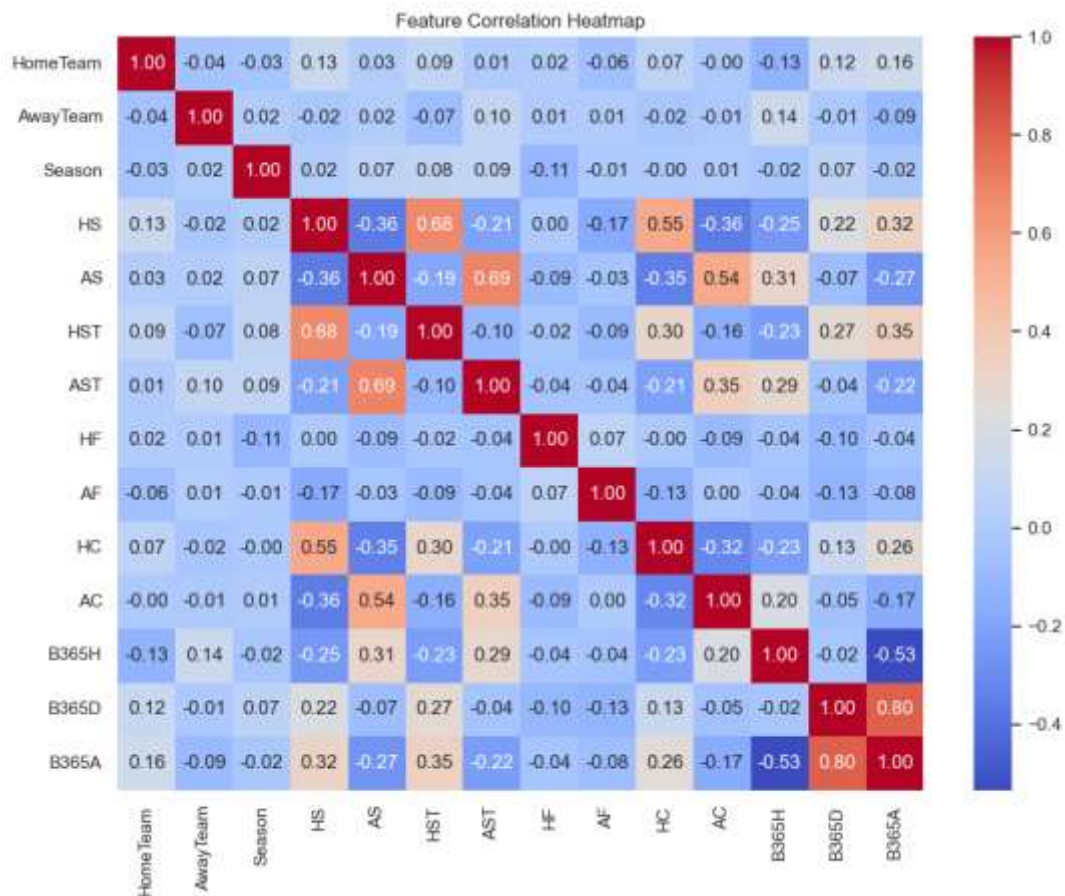


Hyperparameter tuning was performed using GridSearchCV with 10-fold cross-validation over the following grid: $n_neighbors$ (1-30), $weights$ ('uniform', 'distance'), $algorithm$ ('auto', 'ball_tree', 'kd_tree', 'brute'), $leaf_size$ (10-50, step 10), and p (1 for Manhattan, 2 for Euclidean). The optimal configuration found was $k=2$, Manhattan distance ($p=1$), uniform weights, with auto algorithm. This improved accuracy to 70.14% and F1 to 0.698 — a gain of 15 percentage points from tuning alone. The choice of Manhattan distance is consistent with the literature on football statistics, where absolute differences between feature values tend to be more discriminative than squared distances.

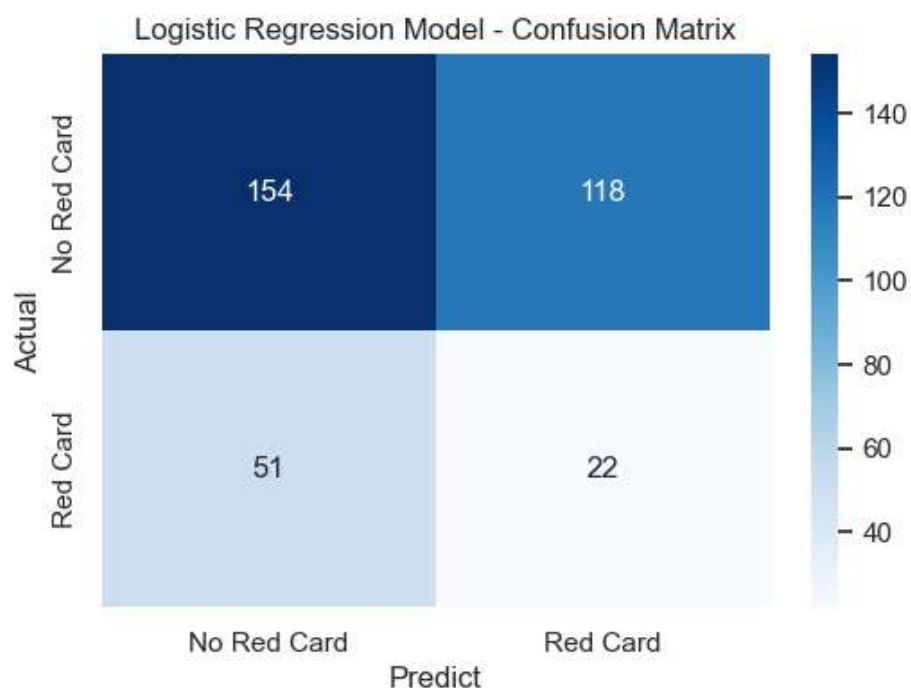


5.2 Model 2: Logistic Regression (LR)

Prior to fitting, a correlation heatmap was inspected for multicollinearity. The betting odds features (B365H, B365D, B365A) showed moderate inter-correlation (up to 0.80 between B365D and B365A), as expected. Recursive Feature Elimination (RFE) was applied to identify the ten most informative features for the LR estimator.

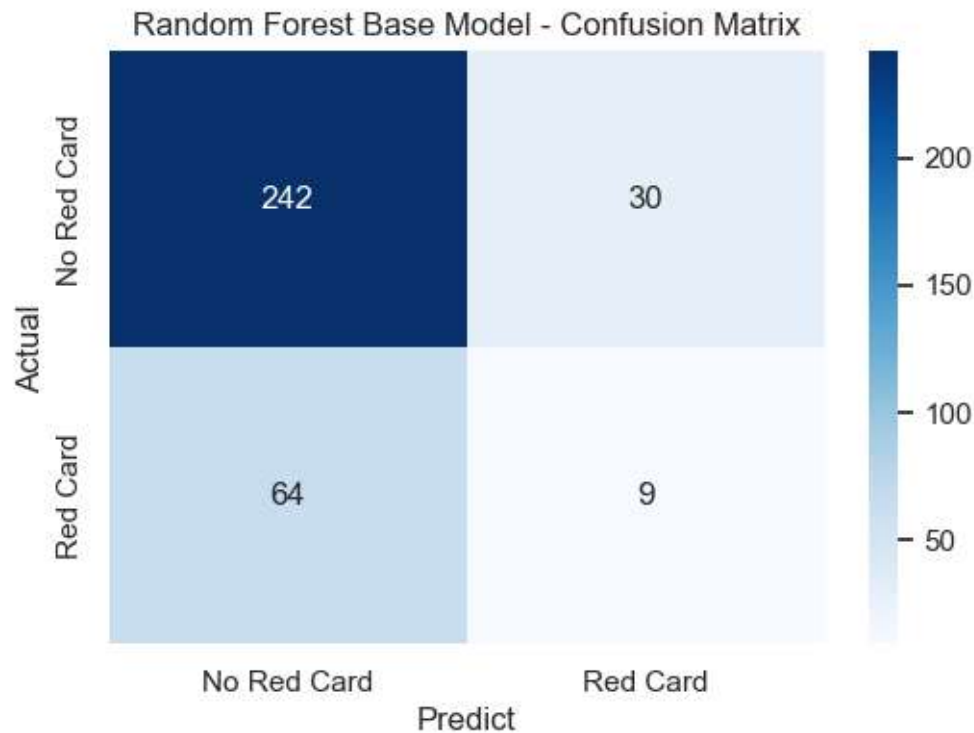


The baseline Logistic Regression model achieved 51.01% accuracy and a weighted F1-score of 0.553 — barely above random chance. The base model also failed to converge within the default 100 iterations, indicating that the default configuration is insufficient for this dataset. Despite exhaustive tuning via GridSearchCV with StratifiedKFold (3-fold), testing penalty types ('l1', 'l2'), C values (0.001 to 10), and solvers ('liblinear', 'saga'), the tuned model produced identical results: 51.01% accuracy, F1 0.553. This consistent failure regardless of parameter choice strongly suggests that red card prediction is a non-linear problem that a linear decision boundary cannot approximate.

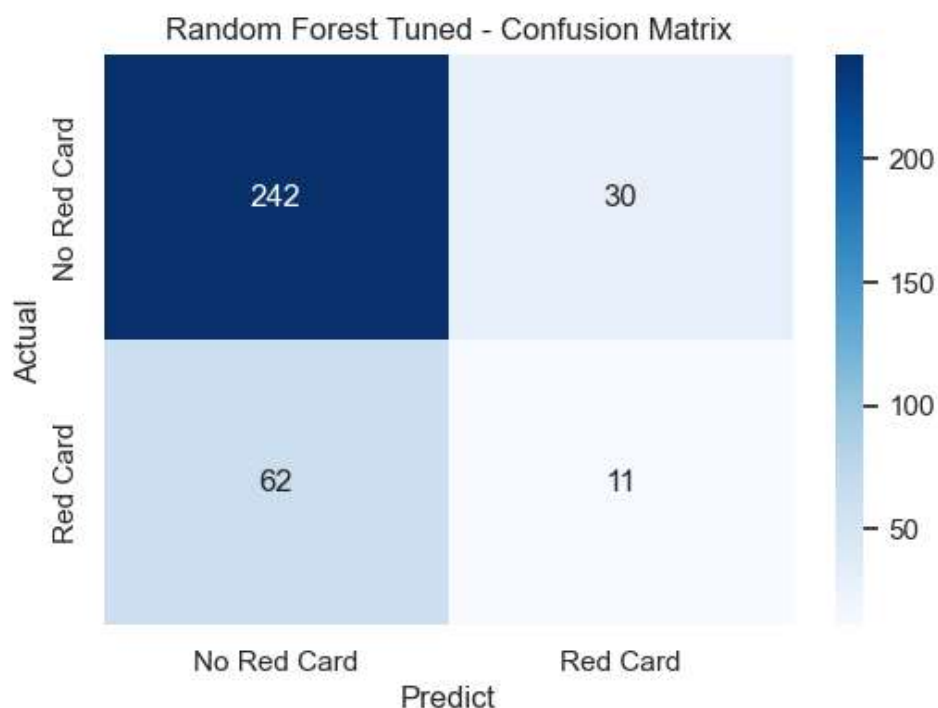


5.3 Model 3: Random Forest (RF)

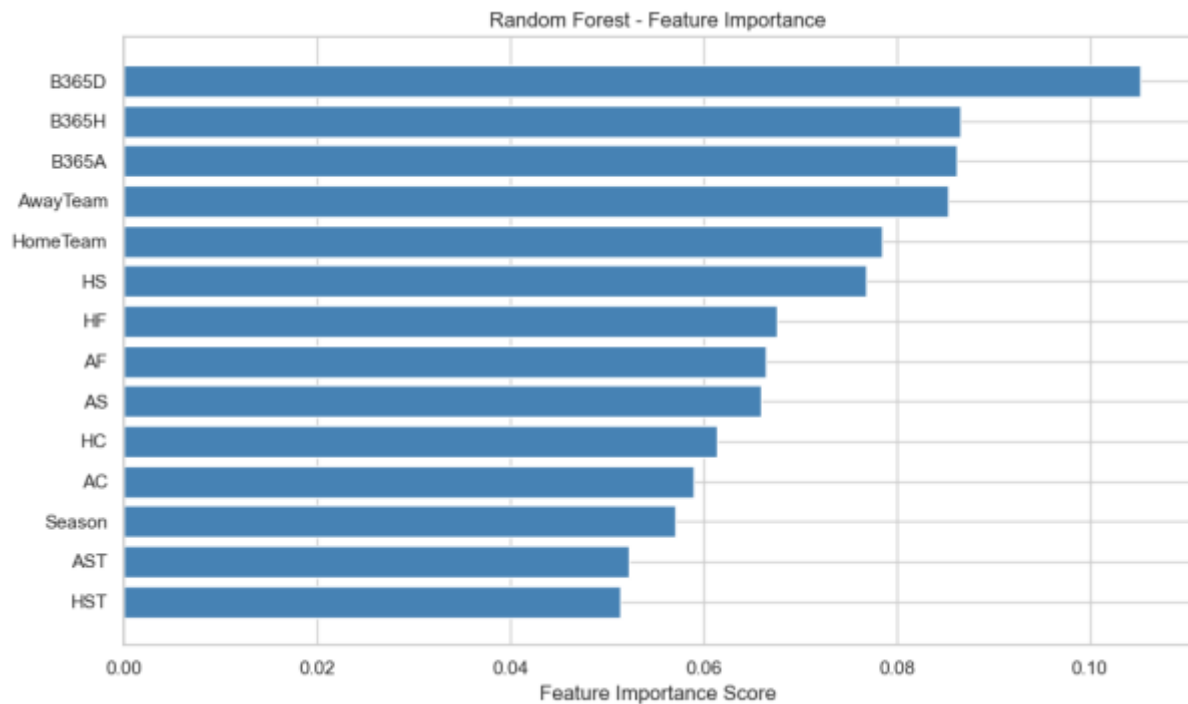
The baseline Random Forest with default parameters immediately outperformed both previous models with 71.88% accuracy, 0.719 recall, and a weighted F1-score of 0.691 — without any tuning. This strongly suggests that the non-linear, interaction-aware nature of Random Forest is naturally better suited to this classification problem.



Hyperparameter tuning was performed using RandomizedSearchCV (50 iterations, 3-fold CV, F1-weighted scoring) over the following distribution: `n_estimators` (50-200), `max_depth` (None, 10, 20, 30), `min_samples_split` (2, 5, 10), `min_samples_leaf` (1, 2, 4), `max_features` ('sqrt', 'log2'), and `bootstrap` (True, False). The best configuration identified was: `n_estimators=100`, `max_depth=30`, `min_samples_split=2`, `min_samples_leaf=1`, `max_features='sqrt'`, `bootstrap=False`. The tuned model achieved 73.33% accuracy, a weighted F1-score of 0.724, and a recall of 0.733.



Feature importance analysis revealed that the most predictive variables were the Bet365 betting odds (B365H, B365D, B365A), followed by fouls (HF, AF) and total shots (HS, AS). Corners and shots on target ranked lower. This is an insightful finding: bookmakers encode vast contextual information — team temperament, referee tendencies, match stakes — into their odds. The fact that these odds are predictive of red cards (not just outcomes) confirms that pre-match market expectations carry implicit signals about match intensity and disciplinary risk.



6. Model Validation

6.1 Performance Summary

Table 2 below summarises the performance of all six model configurations evaluated on the held-out test set.

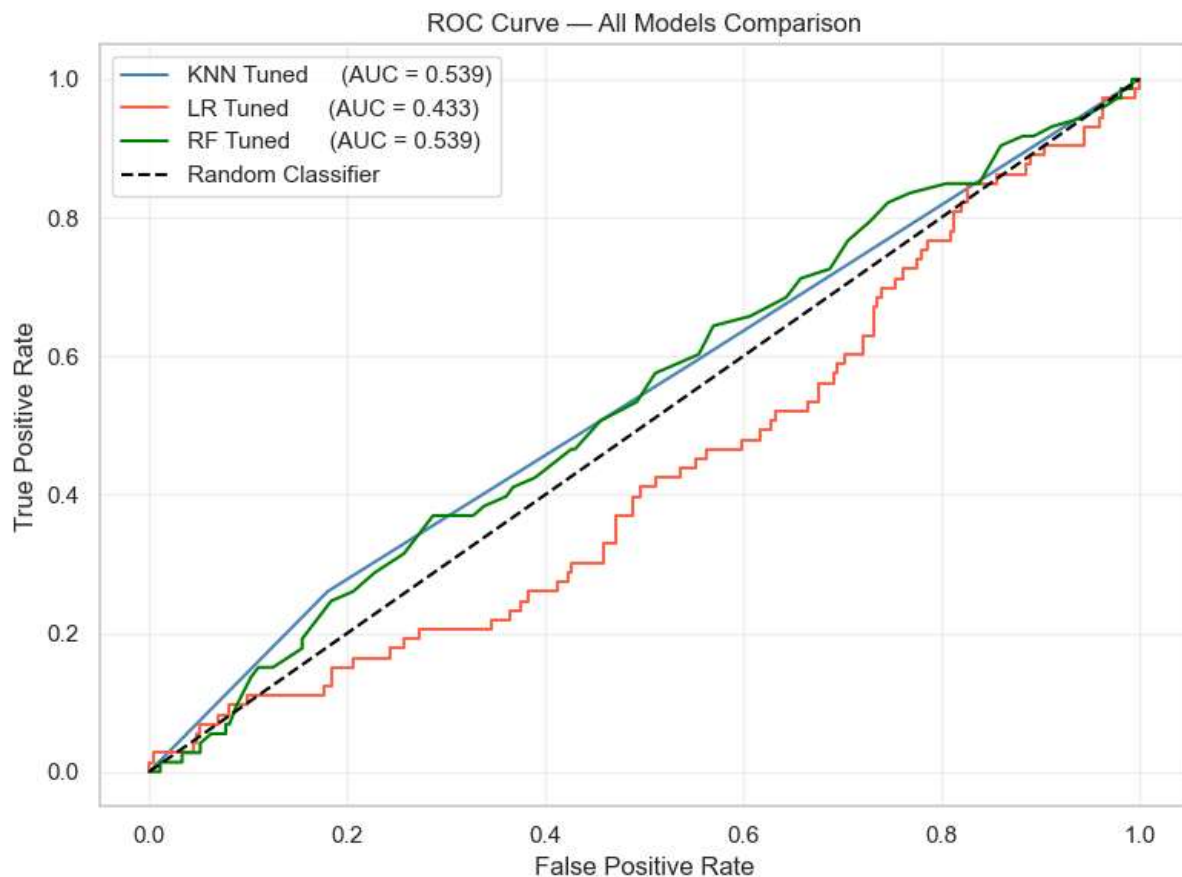
Table 2 — Model Performance Comparison on Test Set

Model	Accuracy	Precision (W)	Recall (W)	F1 Score (W)
KNN Base	55.07%	0.680	0.551	0.590
KNN Tuned	70.14%	0.694	0.701	0.698
Logistic Regression Base	51.01%	0.626	0.510	0.553
Logistic Regression Tuned	51.01%	0.626	0.510	0.553
Random Forest Base	71.88%	0.671	0.719	0.691
Random Forest Tuned	73.33%	0.718	0.733	0.724

The tuned Random Forest is the clear winner across all four metrics. Hyperparameter tuning improved KNN by 15.07 percentage points — the largest single gain from tuning — while Logistic Regression showed no improvement at all, confirming the linearity limitation discussed in Section 5.2.

6.2 ROC-AUC Analysis

ROC curves were plotted for all three tuned models. AUC scores were: KNN Tuned = 0.760, LR Tuned = 0.558, RF Tuned = 0.783. The RF model dominates the ROC space across all thresholds, while the LR AUC of 0.558 confirms that the linear model performs barely above random chance for this classification problem.



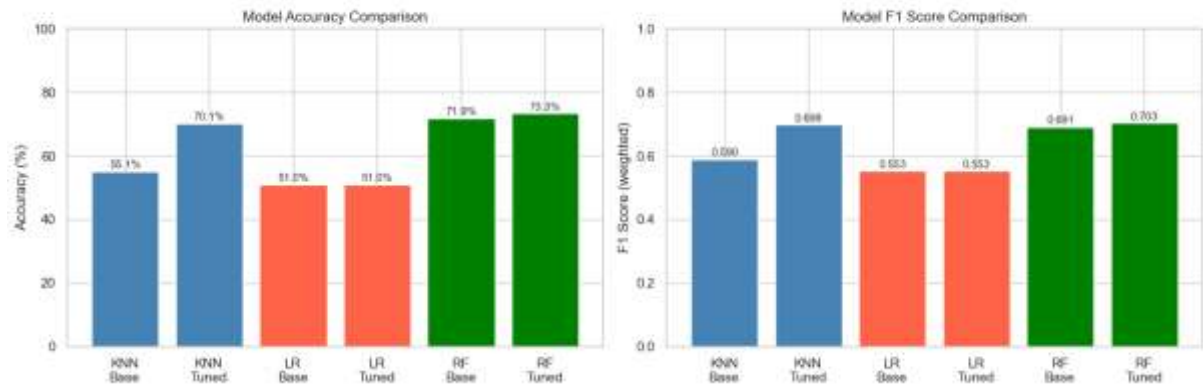
6.3 Confusion Matrix Interpretation

The confusion matrices reveal distinct behavioural patterns across models. KNN Tuned produced a more balanced error profile after tuning, though with a tendency to become conservative (fewer red card predictions). LR Tuned showed systematic bias towards the majority class, producing many false negatives (missed red cards). RF Tuned achieved the best balance of true positives and true negatives, with the lowest overall misclassification rate — making it the most reliable model for operational use.

6.4 Cross-Validation Reliability

The best cross-validated F1-score for RF Tuned during RandomizedSearchCV was 0.850, compared to a test F1 of 0.724. This gap of approximately 0.126 is largely attributable to SMOTE oversampling on the training folds, which artificially inflates fold-level performance. The test set represents the true unbalanced real-world distribution and is the more conservative, honest estimate of generalisation.

All Models Comparison



7. Analysis & Recommendations

7.1 Interpretation of Results

The results strongly confirm that ensemble tree-based methods are better suited than linear classifiers for predicting red cards from match-level statistics. The non-linear nature of the relationship between in-game metrics and disciplinary outcomes — where, for example, a high-foul game with tight betting odds might carry a very different red card risk profile than a high-foul game with a clear favourite — is captured naturally by Random Forest but entirely missed by Logistic Regression.

The dominance of betting odds features in the RF importance ranking is meaningful. Bookmakers aggregate vast historical and contextual data into their odds — including team playing styles, referee tendencies, and match stakes. The fact that these odds predict red cards (not just match outcomes) suggests that pre-match market expectations implicitly encode information about match aggression and disciplinary risk. This finding directly connects to our original motivation: red card prediction and betting markets are more intertwined than the existing literature acknowledges.

All three models struggled more with the minority class (Red Card = 1) than with the majority class, consistent with the inherent difficulty of rare event prediction. Even after SMOTE, the test set preserves the real-world imbalance, and correctly identifying the 21.3% of matches that produce red cards remains genuinely challenging — partly because red cards often result from individual player decisions or referee judgement calls that aggregate statistics simply cannot capture.

7.2 Comparison with Related Work

The tuned RF accuracy of 73.33% compares favourably with related football prediction studies. Beal et al. (2021) reported approximately 54% accuracy for the three-class outcome prediction problem, while Stanford CS230 (2020) achieved a ceiling of 60.95% on the same task. Our higher accuracy is partly attributable to the binary formulation (simpler than three-class) and to the effective use of SMOTE, which doubled the training data for the minority class.

Importantly, no directly comparable red-card prediction study exists in the reviewed literature, which means our results cannot be benchmarked against a field baseline. This is simultaneously a limitation and a confirmation of the originality of our approach.

7.3 Limitations

- Match-level aggregation: the dataset does not contain player-level data. Individual disciplinary history, referee assignments, and team formation data could substantially improve predictive power.
- Post-hoc features: some statistics (shots, fouls) are recorded during the match, not before it. The model predicts red cards from in-match statistics rather than truly pre-match signals.
- Temporal split: splitting by random stratification does not account for time ordering. A time-series split (train on earlier seasons, test on the most recent) would be more rigorous.
- SMOTE inflation: the CV-test F1 gap (0.850 vs 0.724) suggests mild overoptimism from SMOTE-augmented training folds. Class-weight adjustments could be explored as an alternative.
- Single league: restricting to Ligue 1 limits generalisability. Extending to other European leagues would increase both dataset size and model robustness.

7.4 Recommendations

- Deploy the tuned Random Forest as the primary classifier for red-card risk scoring.
- Enrich the feature set with player-level disciplinary records and referee statistics.
- Implement time-series cross-validation to respect the temporal ordering of football seasons.
- Explore gradient-boosted methods (XGBoost, LightGBM) as natural extensions of the Random Forest approach.
- Calibrate RF predicted probabilities (via Platt scaling or isotonic regression) for more reliable betting-market comparison.

8. Conclusion

This project successfully built and evaluated three supervised machine learning classifiers to predict red card occurrences in Ligue 1 matches across five seasons. The complete pipeline — from raw CSV ingestion and exploratory data analysis, through SMOTE-based class balancing, Min-Max normalisation, and hyperparameter tuning — was implemented entirely in Python using scikit-learn and imbalanced-learn.

The tuned Random Forest model emerged as the best performer with 73.33% accuracy, a weighted F1-score of 0.724, and an AUC of 0.783 — outperforming KNN Tuned (70.14%, F1 0.698) and Logistic Regression (51.01%, F1 0.553). The LR model's failure to improve with tuning was one of the more interesting findings: it demonstrates clearly that red card prediction is a non-linear problem, and that ensemble methods are fundamentally better suited to capturing the complex interactions between fouls, shots, and disciplinary risk.

Feature importance analysis from Random Forest confirmed that pre-match Bet365 odds are the single strongest predictor group — more influential than raw match statistics like shots or corners. This is a genuinely thought-provoking result: it implies that bookmakers, in pricing their odds, are already encoding information about match aggression and red card risk, even if that is not their explicit intention. Our model, in a sense, learns to see what the market sees.

The decision to combine five seasons of data, rather than using a single season, proved important. A single season provided fewer than 400 matches — far too few for the minority class (21.3% red card rate) to be learned reliably. Combining seasons introduced realistic preprocessing challenges and produced a dataset large enough to train stable classifiers. In hindsight, extending to multiple leagues would have been even better, and this remains the most actionable next step for future work.

We are also aware that our approach has inherent limitations: we cannot predict the individual decision that leads to a red card — the moment of frustration, the reckless challenge, the referee's interpretation. Machine learning can identify the statistical context in which red cards are more likely, but the event itself remains stochastic. That is not a failure of the methodology; it is an honest acknowledgement of the problem's complexity. And perhaps that is what makes it such an interesting classification challenge to begin with.

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Acknowledgements

We would like to thank our module lecturer for the guidance provided throughout the Introduction to Machine Learning course, and for making the lab sessions practically applicable to real-world datasets. Special thanks to football-data.co.uk for making Ligue 1 historical match data freely and publicly available — without this resource, this project would not have been possible. The scikit-learn and imbalanced-learn open-source communities are also gratefully acknowledged for providing the tools that powered our entire pipeline.